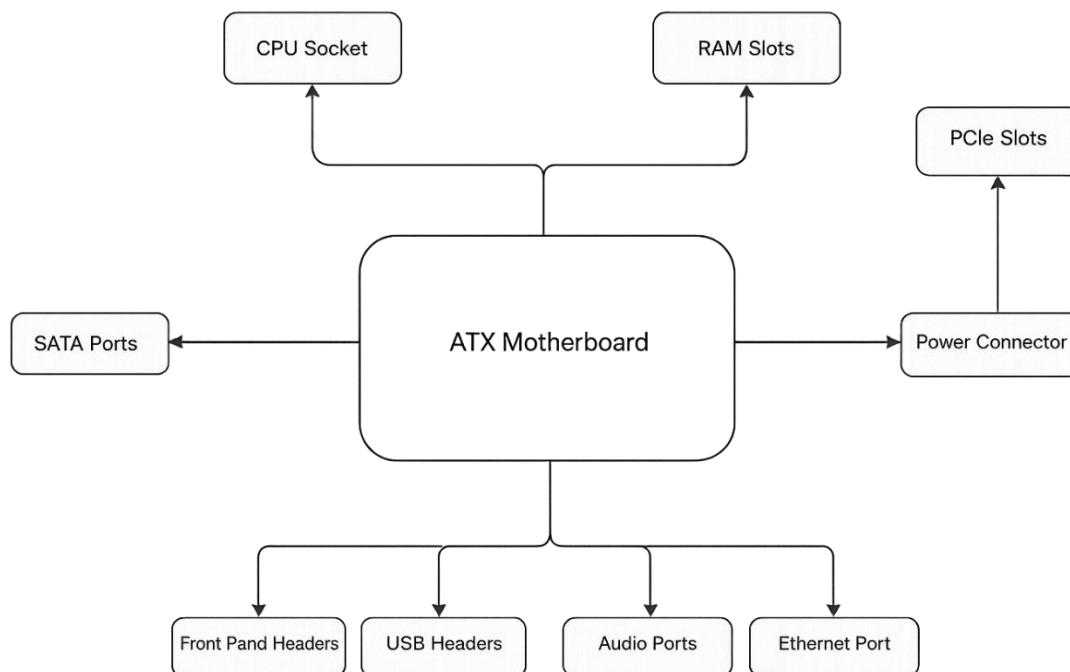


Motherboard and CPU Theory :- Types of motherboards and CPU socket compatibility. Function of the CPU and chipset :-

Ans. :-

Tasks 1 :- Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



Tasks 2 :- Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

LGA1700 :- Intel - Intel Core i5-13600K ,

A modern socket used for Intel 12th, 13th, and 14th Generation Core processors. Supports DDR4 and DDR5 memory.

AM5 :- AMD - AMD Ryzen 7 7700X ,

AMD's latest desktop socket designed for Ryzen 7000 and newer processors. Supports DDR5 memory and PCIe 5.0.

TR4 :- AMD - AMD Ryzen Threadripper 2950X,

A high-end desktop (HEDT) socket used for powerful Ryzen Threadripper processors intended for professional workloads.

Tasks 3 :- Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU (Central Processing Unit) :-

The CPU is the main processor of a computer and is often called the "brain of the computer." Its primary function is to execute instructions, perform calculations, and control the operation of software and hardware. Every task you perform on a computer.

Chipset :-

The chipset is a group of integrated circuits on the motherboard that manages communication between the CPU and other hardware components. It acts like a traffic controller, ensuring that data moves smoothly between different parts of the computer.

the CPU and Chipset Work Together :-

1. You click the Spotify icon.
2. The CPU receives the command to open the application.

3. The chipset helps retrieve Spotify files from the SSD/HDD and loads them into RAM.
4. The CPU processes the instructions and starts Spotify.
5. The chipset manages communication with the internet and audio devices.

Tasks 4 :- Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

